

Ratio & Proportion

Direct & Inverse

Mathematics Lesson

Learning Objectives

By the end of this lesson, students will be able to:

- Understand and explain the concept of a ratio and how to express it
- Identify and set up proportions using the cross-multiplication method
- Distinguish between direct proportion and inverse proportion
- Solve real-world word problems using both types of proportion
- Apply ratio and proportion to daily-life contexts (recipes, maps, sharing)

Lesson Overview

Topic	Key Concept	Method
Ratio	Comparing two quantities	$a : b$ or a/b
Proportion	Two equal ratios	$a:b = c:d$
Direct Proportion	Both go up or down together	$x/y = x/y$
Inverse Proportion	One up, the other down	$x \times y = x \times y$

■ 1. What is a Ratio?

A **ratio** compares two or more quantities and shows how much of one thing there is compared to another. Ratios are written in three equivalent ways:

Form	Example	Read as
Using a colon	3 : 2	"3 to 2"
Using the word "to"	3 to 2	"3 to 2"
As a fraction	3/2	"3 over 2"

Example: 3 apples and 2 oranges

Ratio of apples to oranges = **3 : 2**

Ratio of oranges to apples = **2 : 3**

Note: Order matters! Always check which quantity is mentioned first.

■■ 2. Understanding Proportion

A **proportion** states that two ratios are equal. If **a : b = c : d**, we say these four numbers are in proportion.

Cross-multiplication rule: **$a \times d = b \times c$**

Example: Is 2 : 4 proportional to 1 : 2?

Check: $2 \times 2 = 4$ and $4 \times 1 = 4$ ✓ Both equal 4 → YES, they are proportional!

■ 3. Direct Proportion

In **direct proportion**, when one quantity increases, the other also increases by the same factor. When one decreases, the other decreases too. The ratio between them stays constant.

Formula: $x_1 / y_1 = x_2 / y_2$

Or equivalently: $x_1 \times y_2 = x_2 \times y_1$

Worked Example: If 4 pencils cost ■20, how much do 6 pencils cost?

1	Set up the proportion: pencils/cost is constant $\rightarrow 4/20 = 6/x$
2	Cross multiply: $4 \times x = 20 \times 6$
3	Solve: $4x = 120$, so $x = 120 \div 4$
4	Answer: $x = \text{■}30$ 6 pencils cost ■30

Real-world examples of Direct Proportion:

- More pencils bought $\rightarrow$ higher total cost
- More hours worked $\rightarrow$ more salary earned
- More km driven $\rightarrow$ more fuel consumed
- More flour used $\rightarrow$ more loaves baked
- Larger recipe servings $\rightarrow$ more ingredients needed

■ 4. Inverse Proportion

In **inverse proportion**, when one quantity increases, the other *decreases* so that their **product remains constant**. Think: more workers → less time.

Formula: $x_1 \times y_1 = x_2 \times y_2 = \text{constant}$

The product of the two quantities is always the same.

Worked Example: 4 workers finish a job in 12 hours. How long will 6 workers take?

1	Identify constant: Workers $\times$ Hours = constant
2	Calculate: $4 \times 12 = 48$ worker-hours
3	Set up: $6 \times ? = 48$
4	Solve: Hours = $48 \div 6 = 8$ hours
5	Answer: 6 workers take 8 hours

Real-world examples of Inverse Proportion:

- More workers on a task → fewer hours needed
- Cycling faster → reach school in less time
- More taps filling a tank → less time to fill it
- More passengers sharing a cab → less fare per person
- Higher speed → shorter travel time (same distance)

5. Direct vs Inverse — Key Differences

Direct Proportion	Inverse Proportion
Both quantities change in the same direction	Quantities change in opposite directions
When one increases, the other increases	When one increases, the other decreases
When one decreases, the other decreases	Their product $x \times y$ stays constant
The ratio x/y stays constant	Formula: $x_1 \times y_1 = x_2 \times y_2$
Formula: $x_1/y_1 = x_2/y_2$	Example: More workers = less time needed
Example: More pencils = higher cost	Example: Faster speed = less travel time

6. How to Identify the Type

- Step 1: Read the problem carefully and identify the two changing quantities.
- Step 2: Ask — "If one quantity increases, does the other increase or decrease?"
- Step 3:
 - Same direction (both up or both down) → **Direct Proportion**
 - Opposite directions (one up, other down) → **Inverse Proportion**
- Step 4: Apply the correct formula and solve.

7. Real-World Applications

Situation	Type	Explanation
Cooking a recipe	Direct	Double the servings → double the ingredients. Ratio stays 2:1 or 1:2.
Speed & travel time	Inverse	Cycle faster → reach school sooner. $\text{Speed} \times \text{Time} = \text{constant distance}$.
Sharing money fairly	Ratio	£300 shared in ratio 1:1:1 → £100 each.

Map reading	Direct	Scale 1:1000 means 1 cm on map = 1000 cm (10 m) in real life.
Filling a water tank	Inverse	More taps open → less time to fill the tank. Taps × Time = constant.

8. Practice Problems

Problem 1 [Direct Proportion]

If 5 notebooks cost £150, how much will 8 notebooks cost?

Hint: Find the cost of 1 notebook first, then multiply.

Problem 2 [Inverse Proportion]

3 machines take 10 hours to finish a job. How long will 5 machines take?

Hint: Machines $\times$ Hours = constant total work.

Problem 3 [Ratio Problem]

The ratio of water to juice in a drink is 2 : 3. If you have 10 litres of juice, how much water is needed?

Hint: Water/Juice = 2/3. Set up the proportion.

Problem 4 [Direct Proportion]

A car travels 180 km in 3 hours. How far will it travel in 5 hours at the same speed?

Hint: Distance is directly proportional to time at constant speed.

Problem 5 [Inverse Proportion]

6 workers can paint a fence in 4 days. How many workers are needed to finish it in 3 days?

Hint: Workers $\times$ Days = constant total work.

Problem 6 [Ratio]

Three friends share £600 in the ratio 1 : 2 : 3. How much does each person get?

Hint: Find the value of each part first.

■ 9. Answers with Full Working

Q1

[Direct Proportion]

Step 1: Cost of 1 notebook = ■ $150 \div 5 =$ ■30

Step 2: Cost of 8 notebooks = ■ $30 \times 8 =$ ■240

✓ ■240

Q2

[Inverse Proportion]

Step 1: Total work = 3 machines $\times$ 10 hours = 30 machine-hours

Step 2: Time for 5 machines = $30 \div 5 =$ 6 hours

✓ 6 hours

Q3

[Ratio]

Step 1: Water : Juice = 2 : 3

Step 2: If Juice = 10 L, then Water = $(2/3) \times 10 =$ 6.67 L

✓ 6.67 litres

Q4

[Direct Proportion]

Step 1: Speed = $180 \div 3 =$ 60 km/h

Step 2: Distance in 5 hours = $60 \times 5 =$ 300 km

✓ 300 km

Q5

[Inverse Proportion]

Step 1: Total work = $6 \times 4 =$ 24 worker-days

Step 2: Workers = $24 \div 3 =$ 8 workers

✓ 8 workers

Q6

[Ratio]

Step 1: Total parts = $1 + 2 + 3 =$ 6

Step 2: Each part = ■ $600 \div 6 =$ ■100

Step 3: Shares: $1 \times$ ■100=■100, $2 \times$ ■100=■200, $3 \times$ ■100=■300

✓ ■100 : ■200 : ■300

10. Key Formulas & Summary

Concept	Formula / Rule	Remember
Ratio	$a : b = a/b$	Order matters!
Proportion	$a/b = c/d \rightarrow a \times d = b \times c$	Cross-multiply to check
Direct Proportion	$x_1/y_1 = x_2/y_2$	Same direction change
Inverse Proportion	$x_1 \times y_1 = x_2 \times y_2$	Opposite direction; product constant

Quick Tips for Exams

1. Always **read the problem twice** — identify what is increasing and what is decreasing.
2. For direct proportion: **set up two equal fractions** and cross-multiply.
3. For inverse proportion: **multiply both pairs** (they must be equal).
4. **Show your working** step by step — you earn marks for method, not just the answer.
5. **Check your answer**: does it make sense? (e.g. more workers should take less time)

Keep practising — every ratio is a superpower! ■